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Exploring the intramolecular phosphorylation sites in human Chk2

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ABSTRACT

A comparative biochemical analysis was performed using recombinant human protein kinase Chk2 (checkpoint kinase 2) expressed in bacteria and insect cells. Dephosphorylated, inactive, recombinant human Chk2 could be reactivated in a concentration-dependent manner. Despite distinct time-dependent autophosphorylation kinetics by monitoring the phosphorylation of amino acid residues T68, S19, S33/35, T432, in Chk2 wildtype and Chk2 mutants (T68A, T68D and Q69E) they gave identical specific activities. However, upon gel filtration of Chk2 wildtype and the mutants, only Chk2 wildtype and the T68D mutant led to the formation of a 'pure' dimer; dephosphorylated wildtype Chk2 eluted as a monomer. Transfection of HEK293 cells with Chk2 wildtype and Chk2 mutants in the absence or presence of DNA damage showed significant T68 phosphorylation already in the absence of DNA damaging reagents. Upon DNA damage, phosphorylation of additional Chk2 sites was observed (S19, S33/35). A comparison of $ATM^{+/+}$ and $ATM^{-/-}$ cells with respect to phosphorylation of residues T68, S19, S33/35 in the absence and presence of DNA damage showed in all cases phosphorylation of T68, although signal intensity was increased ca. three-fold after DNA damage. Mass spectrometric analyses of human recombinant Chk2 isolated from bacteria and insect cells showed distinct differences. The number of phosphorylated residues in human recombinant Chk2 isolated from bacteria was 16, whereas in the case of the recombinant human Chk2 from insect cells it was 8. Except for phosphorylated amino acid T378 which was not found in the Chk2 isolated from bacteria, all other phosphorylated residues identified in human Chk2 from insect cells were present also in Chk2 from bacteria.

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1. Introduction

Protein kinase Chk2 (checkpoint kinase 2) belongs to the subfamily of CaMK kinases (calcium/calmodulin-dependent protein kinases) and is activated in response to DNA damage [1]. Chk2 activation is a multistep process, initiated by ataxia telangiectasia mutated (ATM) kinase, which phosphorylates Chk2 on threonine 68 (T68) in a segment of the protein N-terminal to the forkhead homology-associated (FHA) domain [2–4]. The phosphorylated serine-glutamine/threonine-glutamine (SQ/TQ) cluster domain (SCD-segment), harboring T68, binds to the FHA domain of a second Chk2 molecule, promoting dimerisation of the protein and triggering a series of autophosphorylation events involving T383/T387 residues located in the activation segment/T-loop of the kinase domain and S516 in the C-terminus (Fig. 1) [5–8].

Recently, analysis of a partial structure of Chk2 revealed a remarkable dimeric arrangement in which T-loops are exchanged between protomers, to form an active kinase conformation in trans, suggesting a mechanism for dimerisation-driven activation of Chk2 by trans-phosphorylation [9]. Despite the valuable insight into the molecule's structure with respect to the FHA and kinase domains, no structural data are available concerning the amino terminal SQ/TQ domain, harboring the amino acid residues S19, S33, and S35 which upon substitution by alanine show an impaired Chk2 dimerisation accompanied by a failure to inhibit cell growth and, in response to IR, to arrest G1/S progression [10]. The latter findings support the notion that Chk2 activation is not only confined to the phosphorylation of T68 by ATM and subsequent autophosphorylations of T383/387 and S516 residues but one should take into account other phosphorylated serine/threonine amino acid residues recently described in the literature, e.g. S62, S73, phosphorylated by Plk3 [11] and S456, the latter which has been associated with a stability function of the Chk2 molecule [12].

Moreover, King et al. identified 11 novel autophosphorylation sites in human recombinant Chk2, expressed in *E. coli*, using mass

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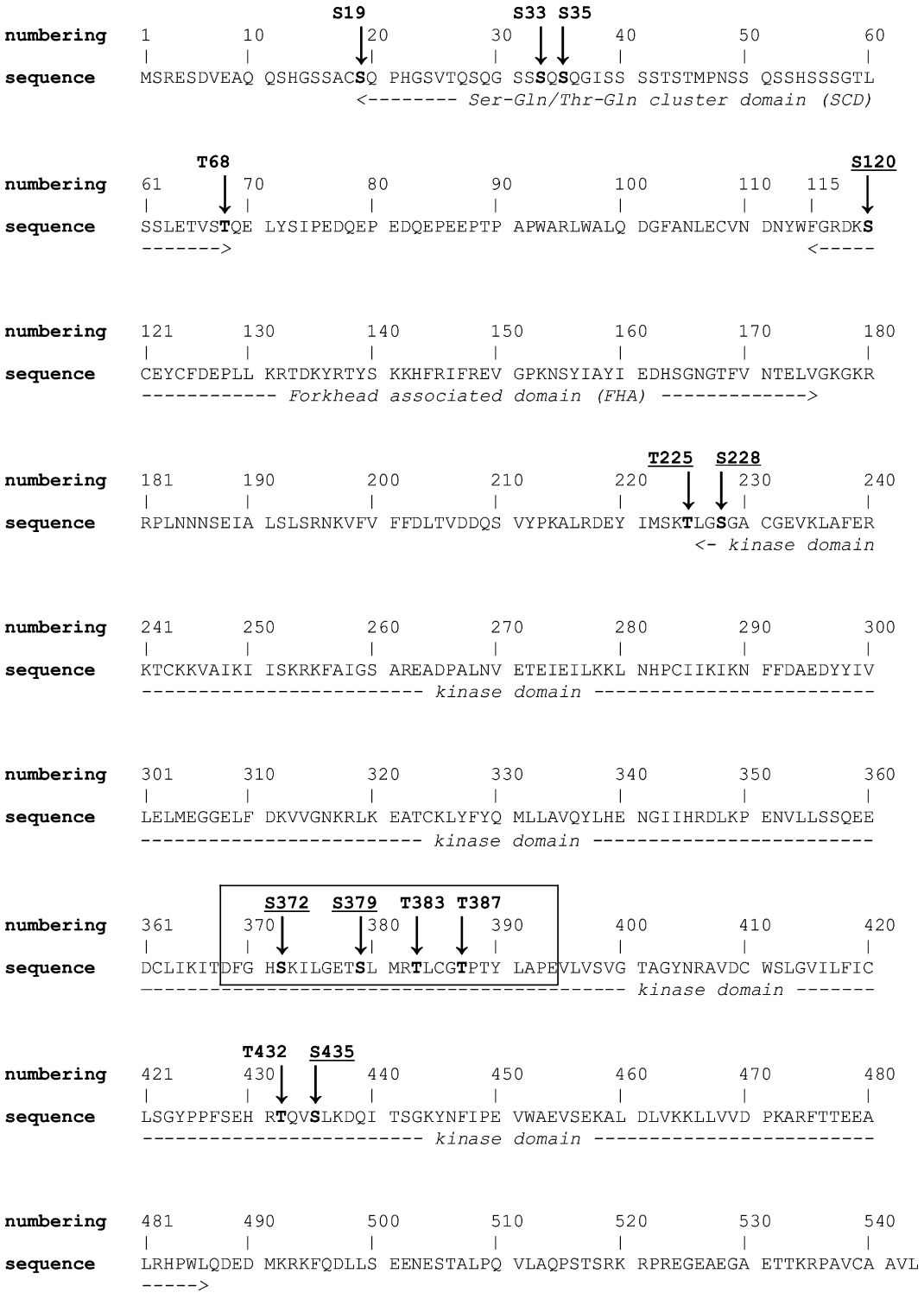


Fig. 1. Chk2 amino acid sequence indicating the main domains and the phosphorylation sites which have been commonly identified by different techniques in agreement with reports from the literature. The phosphorylation sites and their labels are printed with bold characters. The underlined labels mark sites found with mass spectrometry by more than one group; all other sites were detected using phospho-specific antibodies. The activation segment within the kinase domain is indicated with a box.

spectrometry analyses [13] and an additional 17 sites were discovered within the N-terminal SCD domain in a study combining site-directed mutagenesis with mass spectrometry [14]. Hence, the hitherto known Chk2 residues involved in its activation have to be supplemented by new sites [10–12,14].

Upon DNA damage, activated Chk2 phosphorylates a range of downstream targets including Cdc25A and Cdc25B, BRCA1 and p53 ultimately leading to cell cycle arrest, DNA repair or apoptosis [15]. The consensus phosphorylation site in CDC25 (L-X-R-X-X-S/T), one of the first substrates to be identified, has been used to design a substrate peptide (Chktide) for routine screening for Chk2 activity. The presence of this consensus sequence in proteins is used as an indication whether the proteins may serve as possible Chk2 substrates. However, findings from Craig et al. [16] who investigated

the transactivation domain of p53, which does not contain a Chk2 consensus sequence, showed that p53 harbours two docking sites for Chk2, i.e. box-II and box-V, located within the DNA-binding domain. Although located some distance from the T18 and S20 phosphorylation sites within the p53 polypeptide chain, they influence the interaction, such that p53 becomes a substrate for Chk2 [16]. This type of substrate recruitment via remote docking sites resembles the activation of an enzyme by an allosteric effector. It might have widespread significance for Chk2 function since similar mechanisms have been established for other protein kinases such as Cdk2, Pdk1, MAPK, and GSK3 (for further details see Ref. [17]). Moreover, Bjørling-Poulsen et al. [18] have shown that Chk2 specifically interacts with CK2β, the regulatory subunit of protein kinase CK2, and that activation of Chk2 leads to a reduction of this interaction. The presence of the CK2β subunit significantly reduces the Chk2-catalysed phosphorylation of p53 [18].

Hence, the elucidation of the activation mechanism of Chk2 may not only be interesting with respect to the molecule itself under cellular damage but can also be a first step in the understanding how some protein kinases can adopt their structure towards a specific substrate without targeting a specific consensus sequence dogmatic for most of the protein kinases.

In this report we have made efforts to closer validate and identify new phosphorylation sites (i) in Chk2 expressed in mammalian cell cultures, and (ii) in purified recombinant human Chk2 expressed in *E. coli* and *Sf9* cells and provide an up-to-date overview over the phosphorylation sites identified with different techniques in several laboratories. Our results and those available from the literature may be helpful for further investigations concerning the activation mechanism of Chk2.

2. Materials and methods

2.1. Cell culture and transfections

The human embryonic kidney cell line HEK293, which is highly transfectable, was obtained from ATCC, the GM02052F cell line, an ATM-deficient fibroblast culture derived from a 15-year-old girl as well as an age and gender matched normal fibroblast cell culture (GM07532) was obtained from the NIGMS Human Genetic Cell Repository. All three cell lines were cultured in Dulbecco's modified Eagle's medium (DMEM) supplemented with 10% fetal bovine serum (Biochrom AG), penicillin (100 U/ml) and streptomycin (100 μg/ml) in a humidified atmosphere containing 5% CO_2. Transfections were performed using Lipofectamine (Invitrogen) following the manufacturer's recommendations. 46 h after transfection HEK293 cells or the human fibroblasts were incubated with 0.3 μg/ml neocarzinostatin (a gift from Dr. Hiroshi Maeda (Kumamoto University, Japan)) for 2 h.

2.2. Cell lysis and Western blotting

Cells were washed with cold PBS and scraped in cold lysis buffer (50 mM Tris pH 7.5, 150 mM NaCl, 1% Triton X-100, 10% glycerol, 1 mM DTT, 1 mM EDTA, 1 mM Na_3VO_4, 30 mM $NaPP_i$, 10 mM NaF, 100 nM okadaic acid and protease inhibitors (Complete, Roche)) and lysed on ice for 25 min. The lysate was cleared by centrifugation. Proteins were separated by 10% SDS-PAGE and transferred to PVDF membranes (Biorad). Proteins were analysed by probing the membranes with: Polyclonal antiphospho-Chk2 (T68) antibody, antiphospho-Chk2 (S19) antibody, antiphospho-Chk2 (S33/35) antibody, antiphospho-Chk2 (T387) antibody, antiphospho-Chk2 (T432) antibody and polyclonal antiphospho-p53 (S20) antibody (all from Cell Signaling Technology), monoclonal anti-p53 (Calbiochem) and monoclonal anti-Chk2 (Millipore). Protein–antibody complexes were visualized by chemoluminescence following the manufacturer's recommendations (CDP-Star, Applied Biosystems). Western blots were quantified using the Uvi-doc software (Uvitec).

2.3. Plasmid construction and site-directed mutagenesis

Human Chk2 was cloned into the pQE-Trisystem vector (C-terminal His-tag; [18]) for transfection into mammalian cells. For bacterial expression Chk2 was subcloned into pQE31 introducing an N-terminal His-tag (Qiagen) using the restriction enzymes SmaI and HindIII. Site-directed mutagenesis was performed using the QuikChange Site-Directed Mutagenesis Kit (Stratagene).

Lambda PPase was cloned into the pT7-7 vector using the restriction enzymes NdeI and EcoRI. Lambda PPase cDNA was obtained by PCR using lambda DNA digested with HindIII and EcoRI (Fermentas). All clones were verified by sequencing.

2.4. Expression and purification of recombinant proteins

E. coli M15 cells (Qiagen) were transformed with the following plasmids: pQE31-Chk2wt, pQE31-Chk2T68A, pQE31-Chk2T68D or pQE31-Chk2Q69E. Cultures were grown at 37 °C for 2 h and 1.5 h at 22 °C. IPTG was added to a final concentration of 0.1 mM. After growing for an additional 16 h, cells were pelleted by centrifugation and washed in 0.9% NaCl. pT7-7-Lambda PPase was expressed as described above, except the phosphatase was expressed in BL21(DE3) cells and induced with 0.04 mM IPTG. For purification of active Chk2, cells were lysed by sonication in buffer A (50 mM Tris–HCl pH 7.5, 150 mM NaCl, 2 mM DTT, 0.1 mM EGTA, protease inhibitors (Complete, Roche)) and Chk2 was purified by Ni-NTA agarose chromatography (Qiagen). Fractions containing Chk2 were pooled and dialysed against 50 mM Tris–HCl pH 8.0, 150 mM NaCl, 1 mM DTT and subjected to gel filtration (Superdex 200). Inactive Chk2 was purified as described above, except the Chk2 expression pellets were mixed with Lambda PPase pellets and after sonication in buffer A, the lysate was incubated for 30 min at 30 °C with a final concentration of 2 mM $MnCl_2$.

Chk2 was expressed in *Sf9* cells and purified as described earlier [18].

2.5. In vitro kinase and autophosphorylation reactions

The activity of Chk2 was measured in a total volume of 20 μl and contained 200 ng of enzyme, 200 μM synthetic peptide (Chktide KKKVSRSGLYRSPSMPENLNRPR, KinaseDetect), 8 mM MOPS pH 7.5, 0.4 mM EDTA and 18.75 mM $MgCl_2$ and 125 μM [γ-^{32}P]-ATP (specific activity 3000 Ci/mmol, Hartmann Analytic). The reactions were incubated for 10 min at 30 °C and 10 μl were spotted onto P81 phosphocellulose paper and washed with 85 mM phosphoric acid. Radioactivity was measured in a liquid scintillation counter (Canberra-Packard).

For autophosphorylation experiments, Chk2 was incubated at 30 °C for the indicated time points at varying concentrations as indicated in the legends to figures in a reaction mixture containing 8 mM MOPS, 0.4 mM EDTA, 18.75 mM $MgCl_2$ and 125 μM ATP. Where indicated the autophosphorylation reaction contained okadaic acid or a phosphatase inhibitor cocktail (Santa Cruz Biotechnology). Reactions were stopped on ice. The autophosphorylated Chk2 was used for Western blotting or activity testing.

2.6. Chromatographic analysis

50 μg purified recombinant wildtype and mutant Chk2 were analysed by gel filtration on a Superdex 200 PC 3.2/30 column (SMART system, GE Healthcare) equilibrated in 25 mM Tris–HCl pH 8.0, 150 mM NaCl.

2.7. Mass spectrometric analysis of phosphopeptides

Proteins from SDS-PAGE where excised from the gel and submitted to in-gel digestion using standard procedures. The purification of phosphorylated peptides from the resulting peptide mixtures was performed using titanium dioxide (TiO_2) chromatography essentially as described previously [19,20]. After lyophilisation of the in-gel derived peptides they were redissolved in 3 μl 1% SDS and further diluted in 35 μl TiO_2 chromatography loading buffer (80% acetonitrile/5% TFA/1 M glycolic acid) [20]. The solution was applied onto a TiO_2 micro-column and the liquid was gently pressed through the column using a fitted 1 ml plastic syringe. The column was washed with 10 μl loading buffer and 20 μl 80% acetonitrile/1% TFA. The bound phosphopeptides were eluted using ammonia water pH 11.3 (10 μl ammonia (25%) in 490 μl UHQ water).

The purified phosphopeptides were lyophilised prior to nano-liquid chromatography ESI-MSMS as described previously [21].

The nano-liquid chromatography tandem mass spectrometry (nano-LC–MS) was performed using an Orbitrap XL mass spectrometer (Thermo Electron, Bremen, Germany) connected to an EASY nano-LC system (Proxeon Biosystems, Odense, Denmark). The peptides were concentrated on a 1.0 cm pre-column (75 μm inner diameter, 360 μm outer diameter, ReproSil—Pur C18 AQ 3 μm (Dr. Maisch, Germany)). The peptides were eluted from the pre-column using a gradient from 100% phase A (0.1% formic acid aqueous solution) to 40% phase B (0.1% formic acid, 80% acetonitrile) in 100 min at 250 nl/min directly onto a 8 cm analytical column (50 μm inner diameter, 360 μm outer diameter, ReproSil—Pur C18 AQ 3 μm). Peptides were analysed using a data dependent setup in where one MS in the orbitrap at a resolution of 60,000 FWHM was performed and the five highest ions were selected for fragmentation using multi-stage fragmentation for phosphopeptides [22]. The resulting fragment ion spectra were processed (smoothing, background subtraction and centroiding) using the program DTASuperCharge (http://msquant.sourceforge.net/). The processed files were subsequently searched against the recombinant Chk2 protein sequences using an in-house Mascot server (version 2.1) (Matrix science Ltd., London, UK). The search was performed choosing trypsin as enzyme. As variable modification, Oxidation (M), *N*-Acetyl (Protein), Phospho (STY) and IntactPhospho (STY) were chosen. The data were searched with a

peptide mass tolerance of ±5 ppm and a fragment mass tolerance of ±0.6 Da. Maximum 2 missed cleavages were allowed. Phosphopeptides were manually evaluated.

3. Results

3.1. Phosphorylation status of wildtype Chk2 and mutants after transfection of HEK293 cells

In order to closer inspect the importance of the major ATM phosphorylation site (T68) on subsequent intramolecular phosphorylation events within Chk2, leading to its activation, we have performed site-directed mutagenesis. The mutation T68A abolishes the ATM phosphorylation site and the T68D mutation should mimick, at least in part, the ATM-catalysed phosphorylation of T68. By replacing the Q69 residue with a glutamic acid (^{67}STEE^{70}) the ATM consensus sequence (SQ/TQ) is removed.

Fig. 2A shows the phosphorylation status of endogenous Chk2, Chk2-His wildtype and Chk2-His mutants in transfected HEK293 cells in the absence and presence of DNA damage induced by the radiomimetic drug neocarzinostatin (NCS). NCS binds to the minor groove of DNA and induces DNA damage including double strand breaks by free radical mediated fragmentation of the sugar moiety of DNA [23]. Five known phosphorylation sites within Chk2 were monitored for their phosphorylation status using commercially available anti-phospho antibodies directed against P-T68, P-S19, P-S33/35, P-T387, and P-T432. As a loading control the level of Chk2 protein expression was used. The obtained signals for transfected Chk2 were quantified by densitometry and the wildtype signal was set to 100% and compared to the corresponding signals in the mutants.

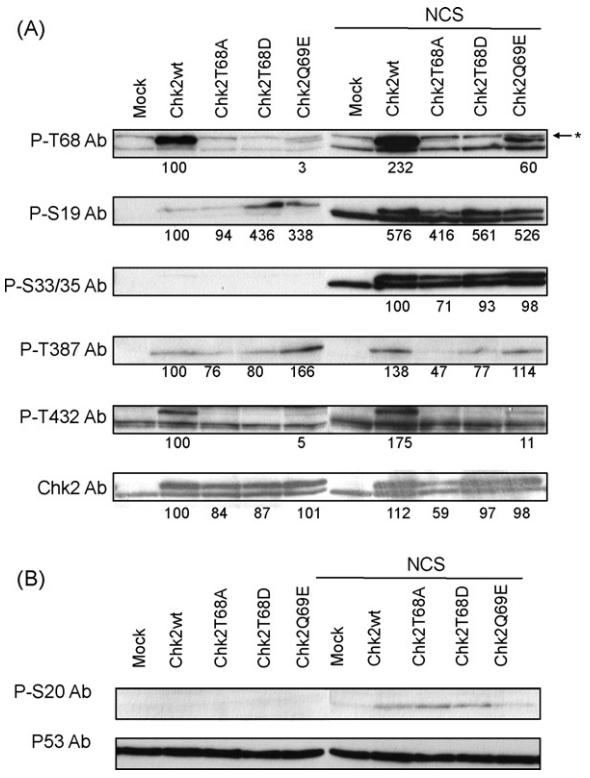


Fig. 2. Phosphorylation status of wildtype Chk2 and mutants in HEK293 cells in the absence and presence of DNA damage. (A) HEK293 cells were transiently transfected with Chk2wt, Chk2T68A, Chk2T68D or Chk2Q69E and either treated with the radiomimetic drug neocarzinostatin (0.3 μg/ml; right panel) or left untreated (left panel). Phosphorylation of T68, S19, S33/35, T387 and T432 was checked by Western blotting using phospho-specific antibodies. Expression of Chk2 was assessed by Western blotting using an antibody directed against Chk2. The lower band is endogenous Chk2, and the upper band the transfected Chk2-His. The asterisk indicates an unspecific band. (B) The same lysates were analysed for S20 phosphorylation of p53 using a phospho-specific antibody. p53 expression was monitored using a p53 antibody. Signal quantification of transfected Chk2-His was performed with the Uvidoc software, and the values are indicated. Untreated cells transfected with wildtype Chk2-His were set to 100%.

Fig. 2A, lane 1 shows mock-transfected cells. Beside a weak phosphorylation of endogenous Chk2 T68 (lower band), no other phosphorylation signals are detected. Upon transfection with wildtype Chk2, a strong phosphorylation of the T68 and T432 residues is observed (lane 2). In addition T387 becomes phosphorylated. S19 phosphorylation is detectable, but very faint and no phosphorylation of amino acids S33/35 is detected.

Transfection with the Chk2T68A mutant results in a phosphorylation pattern (lane 3) resembling that observed in untransfected cells (lane 1), apart from faint phosphorylation signals seen at residues S19 and T387.

Replacing T68 with an aspartic acid residue, and thereby mimicking, at least in part, a phosphorylated amino acid, leads to phosphorylation of S19 and T387 residues (Fig. 2A, lane 4). The phosphorylation of S19 obtained for the T68D mutant is approximately four times stronger than for wildtype Chk2.

A similar phosphorylation pattern is obtained with the Q69E mutant, which abolishes the ATM recognition site (Fig. 2A, lane 5). Here too, as in the case of the T68D mutant, phosphorylation of S19 is approximately three times higher than for wildtype Chk2. Moreover, the signal intensity of P-T387 is significantly increased surpassing the signal strength seen for wildtype Chk2.

When comparing the observed phosphorylation events among each other, it is obvious that T432, in order to become phosphorylated, needs the presence of T68, since it is absent in the mutants. Phosphorylation of S19 and T387 residues undergo only quantitative changes upon T68 mutation. Interestingly, no phosphorylation of S33/35 is detected in the absence of DNA damage.

NCS treatment causes a drastic phosphorylation of residues S19 and S33/35 in comparison to untreated wildtype transfected cells (Fig. 2A, lanes 2 and 7); phosphorylation of T68 and T432 increase only two-fold upon DNA damage.

Transfection with the T68A and T68D mutants does not, as expected, yield any phosphorylation of T68, yet phosphorylation of S19 and S33/35 residues are clearly visible (Fig. 2A, lanes 8 and 9). In the case of the Q69E mutant T68 phosphorylation occurs, albeit the signal strength is only approximately one fourth of that found with wildtype Chk2 (Fig. 2A, lane 7). Moreover, phosphorylation of S19, S33/35, T387 residues and a slight phosphorylation of T432 are observed (Fig. 2A, lane 10).

Since the extent of the phosphorylation of S19 and S33/35 is only slightly less in the Chk2 T68A mutant as compared to Chk2 wildtype, activation of Chk2 activity under DNA damage may depend more on the phosphorylation status of these two residues and less on the phosphorylation of residues T68, T387 and T432.

3.2. Phosphorylation of S20 in endogenous p53 upon transfection with wildtype Chk2 and mutants

Chk2 has been described to phosphorylate S20 in p53 [24]. Therefore, we have used wildtype and mutant Chk2s to investigate their influence on p53 phosphorylation (Fig. 2B). In the absence of neocarzinostatin no p53 S20 phosphorylation is observed (Fig. 2B, lanes 1–5). Mock transfection and subsequent induction of DNA damage does not lead to p53 S20 phosphorylation (Fig. 2B, lane 6). However, upon transfection with wildtype Chk2 (lane 7) but also in the case of the two mutants replacing T68 by either alanine (lane 8) or by aspartic acid (lane 9) phosphorylation of S20 is observed, which suggests that T68 phos-

phorylation might not be important for p53-S20 phosphorylation by Chk2.

Transfection with mutant Q69E leads to a lower phoshorylation level of S20 in p53. This result is kind of a paradox, since this mutant has been shown to yield phosphorylation of the same amino acid residues (apart from the phosphorylation of T432) as found after wildtype Chk2 transfection (Fig. 2A, lanes 7 and 10).

3.3. Phosphorylation status of Chk2 in human primary ATM fibroblasts

Even in the absence of DNA damage an extensive T68 phosphorylation of wildtype Chk2-His was observed suggesting that ATM is not required for T68 phosphorylation. Hence we have investigated phosphorylation of T68, S19 and S33/35 residues in Chk2 from human ATM$^{+/+}$ and ATM$^{-/-}$ fibroblasts (Fig. 3).

In ATM$^{+/+}$ control cells, a distinct phosphorylation of Chk2 T68 can be seen, which increases three-fold upon NCS treatment. This increase is virtually the same as detected upon wildtype Chk2-His transfection of HEK293 cells in the absence and presence of NCS (Fig. 2A, lanes 2 and 7). Moreover, NCS treatment leads to a two-fold increase in S33/35 phosphorylation and *de novo* phosphorylation of S19 (Fig. 3 lane 2).

In ATM$^{-/-}$ cells, phosphorylation of T68 in the control cells (no NCS treatment) was 70% of that found in ATM$^{+/+}$ fibroblasts. Comparing the degree of T68 phosphorylation in ATM$^{+/+}$ fibroblasts with that in ATM$^{-/-}$ cells after DNA damage (Fig. 3 lanes 2 and 4) the phosphorylation signals differ only marginally (taking into account the higher amount of Chk2 in the loading control in lane 4 as compared to lane 2). If we assume that in ATM$^{+/+}$ cells ATM is catalyzing T68 phosphorylation, then a hypothetical kinase in ATM$^{-/-}$ cells is performing at least as well as ATM in targeting T68, e.g. ATR, DNA-PK or TTK/hMps1 [25–27].

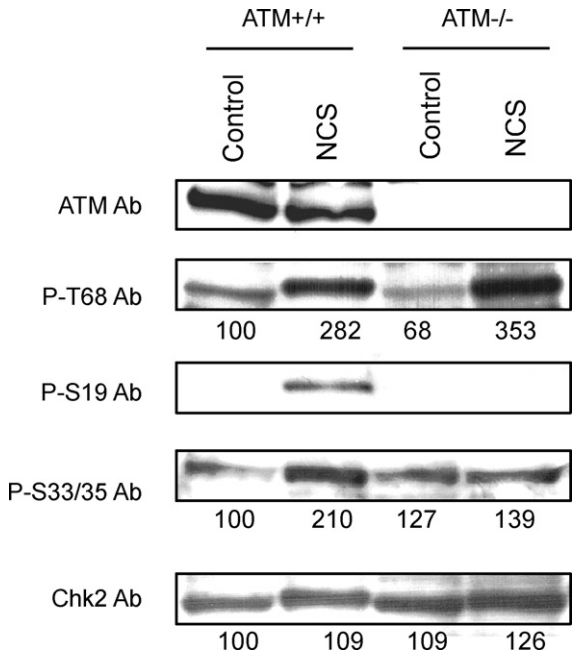


Fig. 3. Phosphorylation status of endogenous Chk2 in human primary fibroblasts in the absence or presence of DNA damage. Human primary fibroblasts (ATM$^{+/+}$ and ATM$^{-/-}$) were treated with neocarzinostatin or left untreated. The phosphorylation status of T68, S19 and S33/35 was checked by Western blotting using phospho-specific antibodies. Expression of ATM and Chk2 was analysed using antibodies directed against ATM and Chk2, respectively. Signal quantification was performed assigning the value 100% to untreated ATM$^{+/+}$ cells.

3.4. Concentration-dependent activation of recombinant human wildtype Chk2 in vitro

In order to further investigate the observation that transfection of wildtype Chk2-His leads to phosphorylation of Chk2 residues T68, S19, T387 and T432 even in the absence of DNA damage, recombinant human wildtype Chk2 was expressed in *E. coli*. Since the enzyme is active upon expression in *E. coli* we have to assume that dimerisation and the obligatory intra- and intermolecular phosphorylation-activation steps must have occurred during expression. Hence, we postulate that when recombinant Chk2 molecules reach a certain concentration, they undergo different steps to activation, implying, that the concentration of Chk2, plays a decisive role for its activation.

To address this hypothesis, we have dephosphorylated active recombinant human Chk2 to inactivate it and then tested for auto-activation activity. Dephosphorylated recombinant human Chk2 was incubated at increasing concentrations (from 0.05 to 0.9 mg/ml) in the absence and presence of phosphatase inhibitors for 1 h (Fig. 4). Activity was measured by incorporation of ^{32}P into the specific peptide substrate. Active recombinant Chk2, as it was isolated from *E. coli*, was set to 100% (bar cluster 1).

The activity of dephosphorylated Chk2 is merely above the detection level, i.e. 5% (bar cluster 2) in comparison to the active enzyme before dephosphorylation. Up to a concentration of 0.3 mg/ml the activity continuously increases reaching a maximum of approximately 60% in comparison to the recombinant

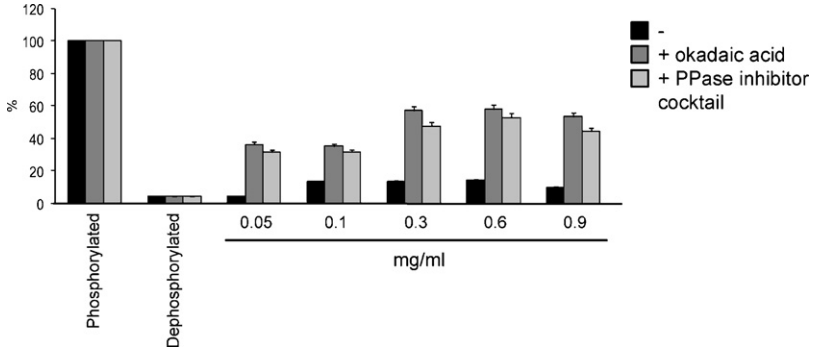


Fig. 4. Autoactivation of dephosphorylated Chk2wt expressed in *E. coli*. Wildtype Chk2 was expressed in bacteria and dephosphorylated by the addition of Lambda PPase prior to purification. Dephosphorylated Chk2 was incubated at increasing concentrations with ATP. The activity was measured by the addition of radioactive ATP and a specific peptide substrate (bar cluster 3–7, black bars). In order to inhibit any remaining lambda PPase activity a phosphatase inhibitor cocktail (light grey bars) or okadaic acid (dark grey bars) was included. As controls dephosphorylated Chk2 (bar cluster 2) as well as Chk2 expressed in *E. coli* and purified in the absence of Lambda PPase (bar cluster 1) were included. The activity of the latter was set to 100%.

Chk2 isolated from *E. coli*, but only in the presence of phosphatase inhibitors. In the absence of phosphatase inhibitors only approximately 15% of the original activity is restored. The much higher Chk2 activity in the presence of phosphatase inhibitors supports the notion that the inactivation of phosphatases, which were not totally removed during the purification steps (Ni-NTA, gel filtration), by phosphatase inhibitors may have been incomplete. However, the results clearly demonstrate a correlation between enzyme concentration and enzyme activity, suggesting, that also in intact cells, when Chk2 concentration is high enough, auto-activation might occur, even in the absence of ATM.

3.5. Time-dependent auto-activation of recombinant Chk2 wildtype, T68 and Q69 mutants

Since Chk2 expressed in *E. coli* is active, autophosphorylation has taken place. In order to examine the effect of mutating the classical ATM T68 phosphorylation site upon autophosphorylation of Chk2 *in vitro*, we have performed time-dependent auto-activation experiments involving wildtype Chk2 and mutants.

Recombinant dephosphorylated Chk2 was monitored for the phosphorylation status of amino acid residues T68, S19, S33/35, and T432 at different times, after incubation with ATP in the presence of a phosphatase inhibitor (okadaic acid). In the case of wildtype Chk2 (Fig. 5A), T68 phosphorylation is apparent after 30 s, slightly increasing up to 1 min and reaching maximum phosphorylation after 5 min. There is no further significant increase seen up to 1 h. Another early phosphorylation is associated with the T387 residue, with maximum phosphorylation after 5 min (results not shown).

S19 and S33/35 phosphorylation are visible after 5 min, reaching a maximum after 40 min. T432 phosphorylation is faintly visible after 10 min, but increases steadily up to 60 min.

These results support the notion of a hierarchical phosphorylation under auto-activation of Chk2.

Auto-activation of the T68A mutant proceeds differently (Fig. 5B). As expected no T68 phosphorylation is observed, yet the onset of phosphorylation of S19, S33/55 and T432 is delayed in comparison to wildtype Chk2 (Fig. 5A). However, in both cases maximum phosphorylation is reached after 40 min, however no phosphorylation of the T387 residue could be detected during the 60 min incubation (data not shown).

Altogether, the T68 mutants (Fig. 5B and C) show a delayed phosphorylation of the residues, S19, S33/35 and T432, hence the T68D mutation could not mimick the phosphorylation of the T68 residue with respect to the timing of the autophosphorylation of the analysed residues. The mutation of the ATM SQ recognition site, by creating a Q69E mutant (Fig. 5D), leads to an 'intermediate' time-dependent phosphorylation pattern as compared to wildtype (A) and the T68 mutants (B and C), Obviously, Chk2 mutated at T68 can achieve a fully active state without prior phosphorylation of T68

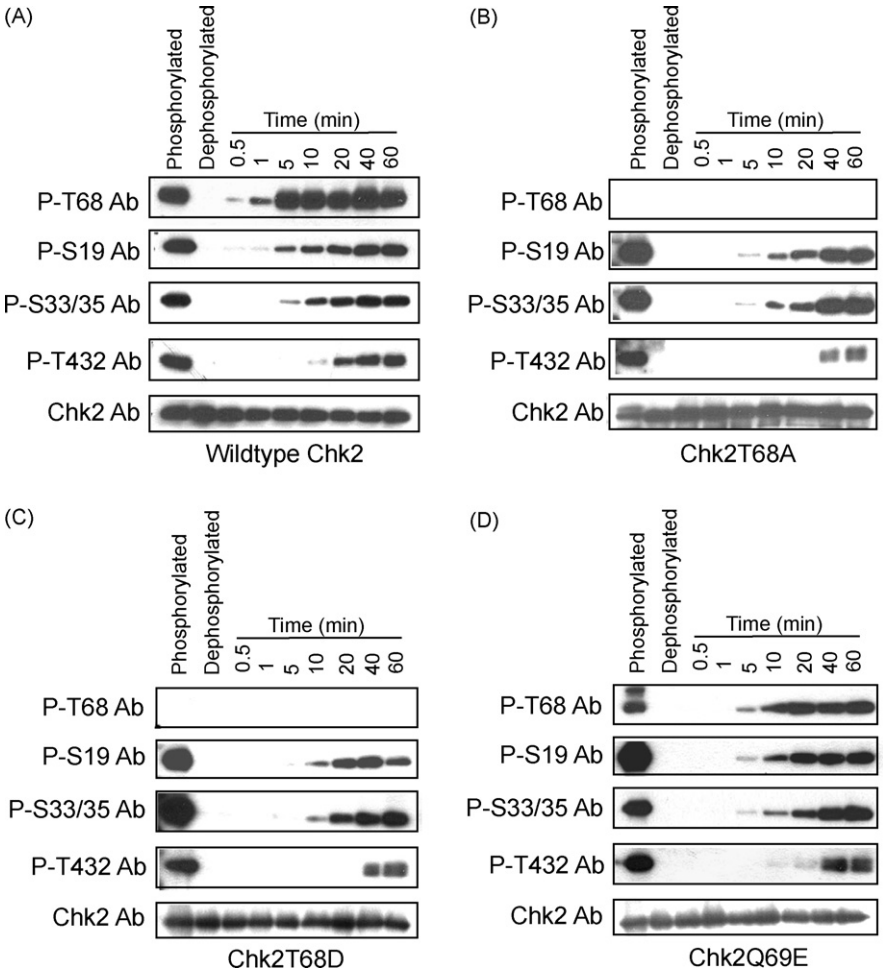


Fig. 5. Time-dependent auto-activation of recombinant wildtype Chk2, Chk2T68A, Chk2T68D and Chk2Q69E. Bacterially expressed wildtype Chk2 and mutants were incubated with Lambda PPase and purified. Purified dephosphorylated Chk2wt (A), Chk2T68A (B), Chk2T68D (C) and Chk2Q69E (D) were incubated with ATP for the indicated times and the phosphorylation status monitored by Western blotting using phospho-specific antibodies against T68, S19, S33/35 and T432. Anti-Chk2 antibody was used to verify equal loading. As a control active Chk2 (Chk2 expressed in bacteria and purified in the absence of Lambda PPase, lane 1) was included.

Table 1
Specific activity of Chk2 wt and mutants

	nmol/min/mg ± S.D
Chk2wt	103.5 ± 8.7
Chk2T68A	96.5 ± 7.3
Chk2T68D	97.0 ± 6.8
Chk2Q69E	101.4 ± 4.6

(Fig. 5B and C). Recombinant Chk2 irrespective whether wildtype or mutated exhibit the same specific activity (Table 1), which support the notion that the enzyme can self-activate during expression in *E. coli*, in the absence of ATM.

3.6. Gel filtration analyses of Chk2 wildtype and mutant forms

In order to answer the question whether the mutations T68A, T68D and Q69E have an influence on the structure of the individual Chk2 variants, the recombinant wildtype Chk2 and mutant forms were analysed by gel filtration at 150 mM NaCl (Fig. 6).

In the phosphorylated state (-PPase) active wildtype Chk2 elutes as a single peak at a position, corresponding to a molecular mass of 160 kDa. Chk2 mutants behave as the wildtype, i.e. the major fraction elutes at 160 kDa. However, unlike the wildtype, a significant part of molecules eluted as >670 kDa aggregates. Attempts to dissociate the high molecular mass material in the presence of 1 M NaCl failed (results not shown) suggesting that once aggregates are formed, this process is irreversible or at least not dissociable at high salt.

Altogether the results obtained suggest that the 160 kDa position of the phosphorylated wildtype and Chk2 mutants represents

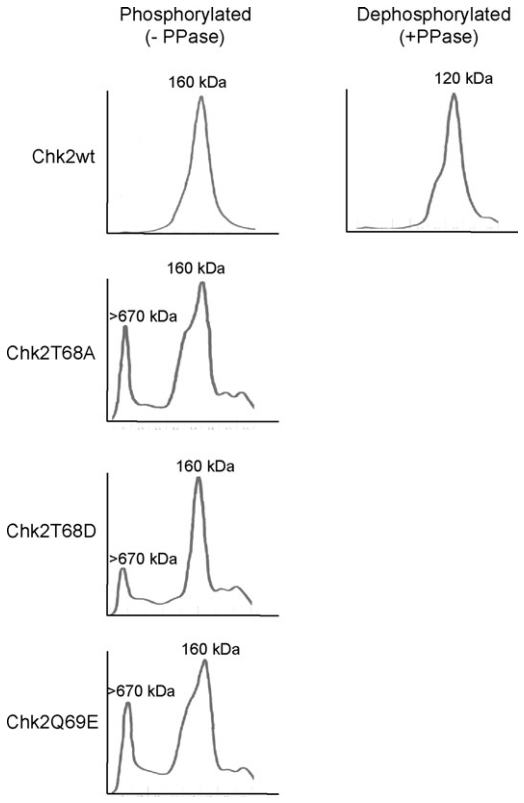


Fig. 6. Analytical gel filtration of recombinant Chk2wt, Chk2T68A, Chk2T68D or ChkwQ69E. 50 μg of recombinant purified Chk2 was analysed by gel filtration chromatography on a Superdex 200 column mounted on a SMART system. The approximate molecular mass of each peak is indicated.

the dimeric form [5]. Upon dephosphorylation the position of the wildtype Chk2 molecule shifts to a position, which corresponds to 120 kDa, i.e. the monomeric form. Although one would expect a monomer to elute at a lower molecular mass, similar experiments performed by Ahn and Prives showed that the monomeric Chk2 eluted at a higher molecular mass than expected, i.e. 114 kDa in comparison to the elution of the homo-dimer, which in their hands eluted at 158 kDa [5]. The similar elution profiles for the monomeric and dimeric Chk2 can be explained by their respective Stokes radii, i.e. 51 Å for the dimer and 42 Å for the monomer [5].

Dephosphorylation of Chk2 leads to a complete loss of enzyme activity (Fig. 4) supporting the notion that the 120 kDa form indeed represents the inactive monomeric Chk2.

3.7. Mass spectrometric analysis

Chk2 activation is associated with a series of phosphorylation events. Phosphorylation occurs intra- and intermolecularly; but also other protein kinases targeting Chk2 have been reported [2,5,6,10–14,24–27]. We have produced recombinant Chk2 in *E. coli* (absence of eukaryotic protein kinases) and in insect cells. Both enzymes were equally active. The recombinant Chk2 proteins were analysed by mass spectrometry for the presence of phosphorylated residues (Fig. 7A).

A total of 16 phosphorylation sites were detected by mass spectrometric analysis of recombinant human Chk2 expressed in *E. coli* (Fig. 7A). From the 8 phosphorylation sites found in human Chk2 expressed in insect cells only one, i.e. T378, is not found in Chk2 expressed in *E. coli*.

Residues S120, S187, S192, S372, T476, T477, S518, T556, and T557 were only found phosphorylated in human Chk2 expressed in *E. coli*. Residues S222, T225, S228, S260, S379, S435 and T441 were found in both recombinant Chk2 molecules. Identification of the phosphorylation sites found in the app. first 100 amino acids by mass spectrometry was, for technical limitations, not possible [13,14].

However, the known three sites located within the first 100 amino acids described in the literature, i.e. S19, S33/35 and T68 as well as the C-terminal T387 and T432 residues were detected using phospho-specific antibodies (Fig. 7B). Interestingly, neither T387 nor T432, which we detected to be phosphorylated in our recombinant proteins, using phospho-specific antibodies were detected by mass spectrometric analysis. Moreover, recombinant human Chk2 expressed in insect cell lacked phosphorylation of T432 (Fig. 7B).

4. Discussion

Numerous studies have demonstrated that ATM phosphorylation of Chk2 following DNA damage occurs predominantly on T68 [2–4]. Hence, these results strongly suggest a role of the FHA domain as a first step in the dimerisation of Chk2. However, the strict requirement of a phospho-dependent binding is circumstantial inasmuch that also phospho-independent FHA domain interactions have been suggested [28]. Moreover, a recent paper based on the crystal structure of a Chk2 fragment comprising amino acid residues 210–531, encompassing the kinase domain (Fig. 1), showed that the Chk2 activation mechanism is more complex than originally thought [9].

In the present paper we have attempted (i) to closer inspect the activation of recombinant Chk2 in the absence of ATM. (ii) To tackle the role of T68 during Chk2 dimerisation by applying analytical gel filtration and the role of T68 in inter- and intramolecular autophosphorylations. The role of T68 was analysed using recombinant wildtype Chk2 and mutants comprising residues T68 and Q69. (iii) Transfecting cells with wildtype Chk2 and the mutant Chk2 constructs (T68A, T68D, Q69E) to monitor the impact on the phos-

(A)

E.coli		Sf9	
Peptide	*Phosphorylated aa*	*Peptide*	*Phosphorylated aa*
DKSCEYCFDEPLLK	S^{120}		
RPLNNNSEIALSLSR	$S^{187} + S^{192}$		
ALRDEYIMSK	S^{222}	ALRDEYIMSK	S^{222}
TLGSGACGEVK	$T^{225} + S^{228}$	TLGSGACGEVK	$T^{225} + S^{228}$
FAIGSAR	S^{260}	FAIGSAR	S^{260}
ITDFGHSK	S^{372}		
ILGETSLMR	S^{379}	ILGETSLMR	$T^{378} + S^{379}$
TQVSLKDQITSGK	$S^{435} + T^{441}$	TQVSLKDQITSGK	$S^{435} + T^{441}$
FTTEEALR	$T^{476} + T^{477}$		
FQDLLSEENESTALPQVLAQPS TSR	S^{518}		
EGEAEGAETTK	$T^{556} + T^{557}$		

(B)

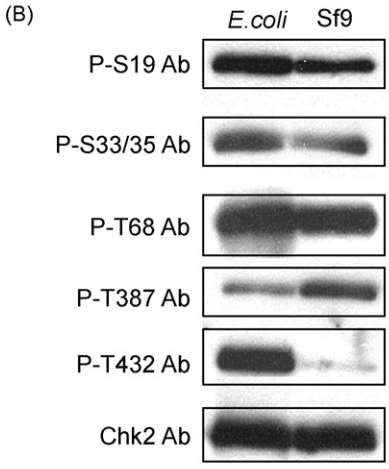


Fig. 7. Phosphorylation pattern of Chk2 expressed in *E. coli* and Sf9 cells. (A) Chk2 was expressed in either *E. coli* or Sf9 cells, purified and used for mass spectrometry analysis. The peptides and the phosphorylated amino acids identified are indicated. (B) Western blot analysis using phospho-specific antibodies directed against T68, S19, S33/35, T387 and T432 residues to determine the phosphorylation status of Chk2 expressed in *E. coli* or *Sf9* cells. The level of Chk2 was used to verify equal protein loading using an anti-Chk2 antibody. Neither of the phosphorylated amino acids assessed by Western blotting were found by mass spectrometric analysis.

phorylation status of S19, S33/35, T387 and T432 residues in the absence and presence of DNA damage.

(i) Expression of highly active recombinant human Chk2 in *E. coli* has been described [6] and the activation was interpreted to result from the high expression levels in bacteria. The results from the concentration-dependent auto-activation of dephosphorylated (inactive) purified recombinant Chk2 (Fig. 4), support the notion that the concentration of Chk2 molecules is an important step in the auto-activation mechanism, despite the absence of ATM or other protein kinases. Although our experimental setup only leads to a ca. 60% restoration of Chk2 activity, there is a clear relationship between enzyme concentration and enzyme activity. Since only in the presence of phosphatase inhibitors a significant restoration of activity was seen, one could argue that the lack of 100% restoration could be due to the presence of contaminating phosphatase activity, originally used to dephosphorylate Chk2. Assuming that phosphatase activity was completely blocked by the presence of phosphatase inhibitors, the lack to fully restore Chk2 activity could be explained by an incomplete intra- and intermolecular phosphorylation.

(ii) Recombinant Chk2 wildtype, Chk2T68A, Chk2T68D and Chk2Q69E were dephosphorylated and subsequently incubated in the presence of ATP up to 1 h (Fig. 5). The results show a hierarchical phosphorylation of the investigated phospho residues T68, S19, S33/35, T432 and T387 (the latter not shown). Remarkably, in a fully phosphorylated state wildtype Chk2 and its mutants all exhibit the same specific activity (Table 1), irrespective of the presence or absence of T68 phosphorylation supporting the notion that a T68-independent FHA domain interaction, as suggested previously [28] also is possible. The gel filtration results show that only Chk2 wildtype and the phosphorylation mimetic Chk2 T68D mutant protein exhibit a "distinct" dimer (apart from the >670 kDa molecular mass material found in the case of the Chk2T68D mutant). The other two mutants (Chk2T68A, Chk2Q69E) show a distinct shoulder in the elution profile close to the position, where the dimer elutes. This material could be trimeric or tetrameric oligomers owing to impairment of the T-loop exchanged dimer described by [9]. The model by Oliver et al. [9] suggests that the transactivation mechanism by T-loop exchange may allow two scenarios: autophosphorylated Chk2 remains as an active T-loop exchanged dimer, or the T-loops fold back to give two conventional kinase domain structure, the latter could interact with another Chk2 molecule and thereby explain our observations during gel filtration analyses. Indeed, the Chk2T68A mutant does not show any phosphorylation of the T387 residue, supporting this notion (data not shown).

(iii) Transfection with wildtype Chk2 of mammalian cells in the absence of DNA damage leads to a drastic phosphorylation of residues T68 and T432, accompanied by moderate phosphorylation of residues S19 and T387 (Fig. 2). However, this incomplete phosphorylation status of Chk2 is not sufficient to catalyze phosphorylation of S20 in p53. Our results show that Chk2 wildtype phosphorylates p53 at S20, only after DNA damage, however phosphorylation of p53 S20 is also seen in the case of the Chk2 T68A mutant, even to an extent as seen

for the wildtype Chk2 (Fig. 2B), supporting the notion that the lack of phosphorylation of residues T68, T387 and T432 in the Chk2T68A mutant (Fig. 2, lane 8) does not compromise Chk2 activity and that other phosphorylation events within Chk2, e.g. at residues S19, S33/35 and very likely many others too, are required to fully activate Chk2.

Since S20 in p53 is not only phosphorylated by Chk2, but also by other protein kinases such as MAPKAPK2, JNK and PLK3 kinase [24] one could speculate that one of these kinases is responsible for the observed phosphorylation of S20 in p53 upon DNA damage. However, since no such phosphorylation is found in mock transfected cells after DNA damage this notion is not convincing and by the same token our results also argue against endogenous Chk2 being responsible for S20 phosphorylation of p53.

In $ATM^{+/+}$ and $ATM^{-/-}$ cells endogenous T68 phosphorylation is present in the absence of DNA damage, which is in agreement with results obtained in HEK293 cells, where also endogenous T68 phosphorylation is detected. Upon DNA damage an increase in T68 phosphorylation occurs in both ATM cell lines. Relating the P-T68 signal intensities observed after DNA damage to the Chk2 expression level, the extent of T68 phosphorylation is very similar, irrespective of the presence or absence of ATM. Phosphorylation of T68 in the absence of ATM kinase could be explained by the action of other kinases, e.g. ATR, DNA-PK or TTK/hMps1 [25–27].

The S19 and S33/35 sites are embedded in SQ clusters and therefore potential target sites for kinases of the phosphatidylinositol 3-kinase family (ATM, ATR and DNA-PK), yet only in the case of $ATM^{+/+}$ fibroblasts increased phosphorylation upon induction of DNA damage is observed. No phosphorylation of S19 can be detected in $ATM^{-/-}$ fibroblasts, whereas a basal S33/35 phosphorylation is observed in $ATM^{-/-}$ cells which are not affected by NCS treatment. Assuming that ATM is responsible for S33/35 phosphorylation in $ATM^{+/+}$ cells, our results support the notion that ATM kinase is the primary responder to IR or radiomimetic agent-induced DNA damage with respect to S19 and S33/35 phosphorylation. However, in the absence of ATM, or in normal cells, expressing ATM, that encounter a high level of IR-induced DNA damage, ATR possibly serves as a backup kinase for ATM but obviously only with respect to T68 phosphorylation (for a review, see Ref. [29]).

King et al. [13] had shown by mass spectrometry analysis, that many more amino acid residues within Chk2 become phosphorylated [13,14], hence we revisited their results in bacteria and moreover, compared these results with recombinant human Chk2 expressed in insect cells. Mass spectrometry analysis showed that twice as many sites were found in Chk2 expressed in bacteria than in Chk2 expressed in insect cells, suggesting that there is a correlation between the high density of expression in bacteria as compared to the expression in insect cells.

Phosphorylation of residues T387 and T432, which we detected in Chk2 expressed in *E. coli* were not detected by mass spectrometric analysis. Interestingly P-T432 was also not detected in human recombinant Chk2 when expressed in insect cells. This is possibly due to the small size of the tryptic peptide covering the T432 site, decreasing the chances for its detection in mass spectrometry analysis. The only phosphorylated residue, i.e. P-T378, which was found only in recombinant human Chk2 expressed in insect cell may be due to an exogenous kinase, since this site previously has been found to be phosphorylated, when transfected Chk2 was purified from HEK293 cells and analysed by mass spectrometry [30].

Table 2 shows an overview of Chk2 phosphorylation sites so far identified. The lack of mass spectrometry data from the N-terminal part of Chk2 was due to the lack of appropriate protease cleavage sites but was recently overcome. The combination of site-directed mutagenesis with mass spectrometry revealed 17 new phosphorylation sites within the N-terminal part of Chk2 (i.e. S12, S15, S24, T26, S39, S40, S41, S42, T43, S44, T45, S50, S52, S55, T65, and S67) in addition to S19, S33/35, S62, T68, and S73 which have already been detected by other methods [13].

Table 2
Phosphorylated amino acid residues in Chk2

	Chk2 (*E. coli*) sites identified by MS in this paper	Chk2 (*E. coli*) sites identified by MS [13]	Chk2 (*Sf*9) sites identified by MS in this paper	Other techniques
S-19				x
S-33/35				x
S-62				x
T-68				x
S-73				x
S-120	x	x		
S-187	x			
S-192	x			
Y-220		x		
S-222	x		x	
S-223		x		
T-225	x	x	x	
S-228	x	x	x	
S-260	x	x	x	
T-272		x		
S-372	x	x		
S-378			x	
S-379	x	x	x	
T383				x
T-387				x
T-432		x		x
S-435	x	x	x	
T441	x		x	
T476	x			
T477	x			
S-516		x		x
T518	x			
T-532		x		
T-556	x			

Comparing our results obtained from recombinant human Chk2 produced in bacteria with those from King et al. there are many sites, which are detected in common, i.e. S120, T225, S228, S372, S379, and S435. Yet, other sites such as Y220, S260, T272, T432, S516, and T532 were only found by [13]. We found sites which were not found by King et al., i.e. S187, S192, S222, T441, T518, T556, and T557. The differences in the phosphorylation sites described above between the two groups may be due to ambiguities in the technique used for phosphopeptide analysis. However, we would like to emphasize that the use of two different expression systems may solidify some of the diverging data.

Taken everything into consideration we conclude:

1. Recombinant Chk2 protein can achieve a fully active state in the absence of exogenous protein kinases in a concentration-dependent manner. Moreover, this auto-activation can also occur in the absence of a T68 residue, yielding a similar specific activity as seen with the wildtype Chk2.
2. Although the presence of a phosphorylatable T68 obviously is not required for the final activity status of Chk2; it is important to initiate proper dimerisation via the FHA domains, a pre-requisite for an active T-loop exchanged dimer (involving residues T383/387 in the kinase domain). The lack of T68 may impair FHA domain dimerisation and thus favor the folding back of T-loops giving rise to two conventional kinase domain structures which may interact with other Chk2 molecules [9].
3. Chk2 becomes phosphorylated at various sites either by exogenous protein kinases or by auto-phosphorylation, both events leading to an active enzyme. Whether a 'proper' dimerisation – involving an intact T68 site – is mandatory *in vivo* to secure an active enzyme has to be resolved in the future.

Note added in proof

While this manuscript was under submission the following paper, reporting data in agreement with our results came to our attention: G. Gabant, A. Lorphelin, N. Nozerand, C. Marchetti, L. Bellanger, A. Dedieu, E. Quéméneur, B. Alpha-Bazin, Autophosphorylated residues involved in the regulation of human Chk2 in vitro, J. Mol. Biol. 380 (2008) 489–503.

Conflict of interest

The authors declare that there are no conflicts of interest.

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